

IBM - VIT- Cybersecurity

ASSIGNMENT - 1

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Analyzing a simple HTTP request using **WIRESHARK**

OBJECTIVE:

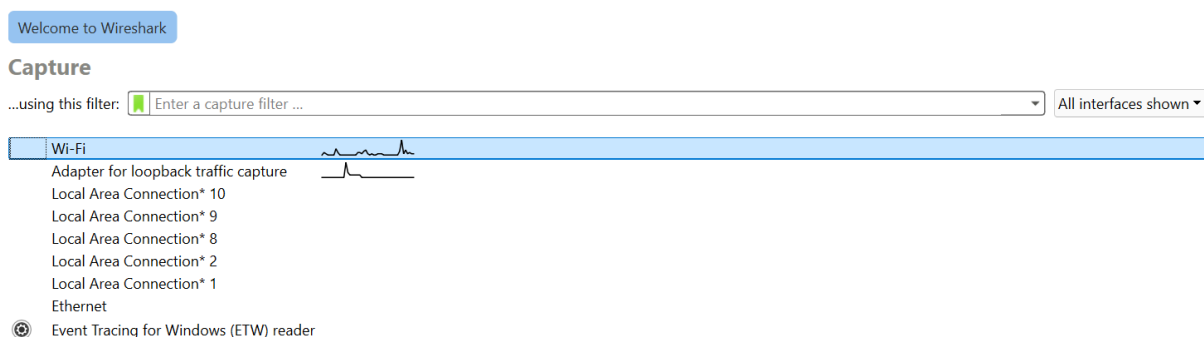
1. Understand how HTTP requests and responses work.
2. Identify source and destination IP addresses.
3. Extract useful information like the host, user- agent, and response codes.

TCP Analysis:

- How many packers are exchanged during the TCP 3-way handshake?
- Check flags in SYN, SYN-ACK, ACK packets.
- What are the sequence and acknowledgment numbers?

WIRESHARK CAPTURE

Launching wireshark : Selecting Wi-Fi from the list of network interfaces and capturing all networks in this interface.



Request and Response for <http://example.com>

Filtering packets: Applying http or tcp.port==80 filter.

No.	Time	Source	Destination	Protocol	Length	Info
368	16.421137	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52918 [FIN, ACK] Seq=1 Ack=1 Win=507 Len=0
369	16.421137	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	584	[TCP Out-Of-Order] 80 → 52918 [PSH, ACK] Seq=4294966787 Ack=1 Win=507 Len=510
370	16.421137	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	584	HTTP/1.0 408 Request Time-out (text/html)
371	16.421137	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52919 [FIN, ACK] Seq=511 Ack=1 Win=507 Len=0
372	16.421268	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52918 → 80 [ACK] Seq=1 Ack=4294966787 Win=255 Len=0
373	16.421553	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52918 → 80 [ACK] Seq=1 Ack=2 Win=254 Len=0
374	16.421615	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52919 → 80 [ACK] Seq=1 Ack=512 Win=254 Len=0
1208	59.026014	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52918 → 80 [FIN, ACK] Seq=1 Ack=2 Win=254 Len=0
1209	59.026106	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52918 → 80 [RST, ACK] Seq=2 Ack=2 Win=0 Len=0
1210	59.026268	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52919 → 80 [FIN, ACK] Seq=1 Ack=512 Win=254 Len=0
1211	59.026321	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52919 → 80 [RST, ACK] Seq=2 Ack=512 Win=0 Len=0
1214	59.061932	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	86	52926 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1217	59.064005	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	86	52927 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1228	59.260333	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52918 [ACK] Seq=2 Ack=2 Win=507 Len=0
1230	59.274945	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52919 [ACK] Seq=512 Ack=2 Win=507 Len=0
1231	59.285030	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	86	80 → 52926 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1300 SACK_PERM WS=128
1232	59.285219	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52926 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1233	59.286022	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	86	52928 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1234	59.286553	2401:4900:2346:4f31...	2600:1406:3a00:21::...	HTTP	500	GET / HTTP/1.1
1235	59.306295	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	86	80 → 52927 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1300 SACK_PERM WS=128

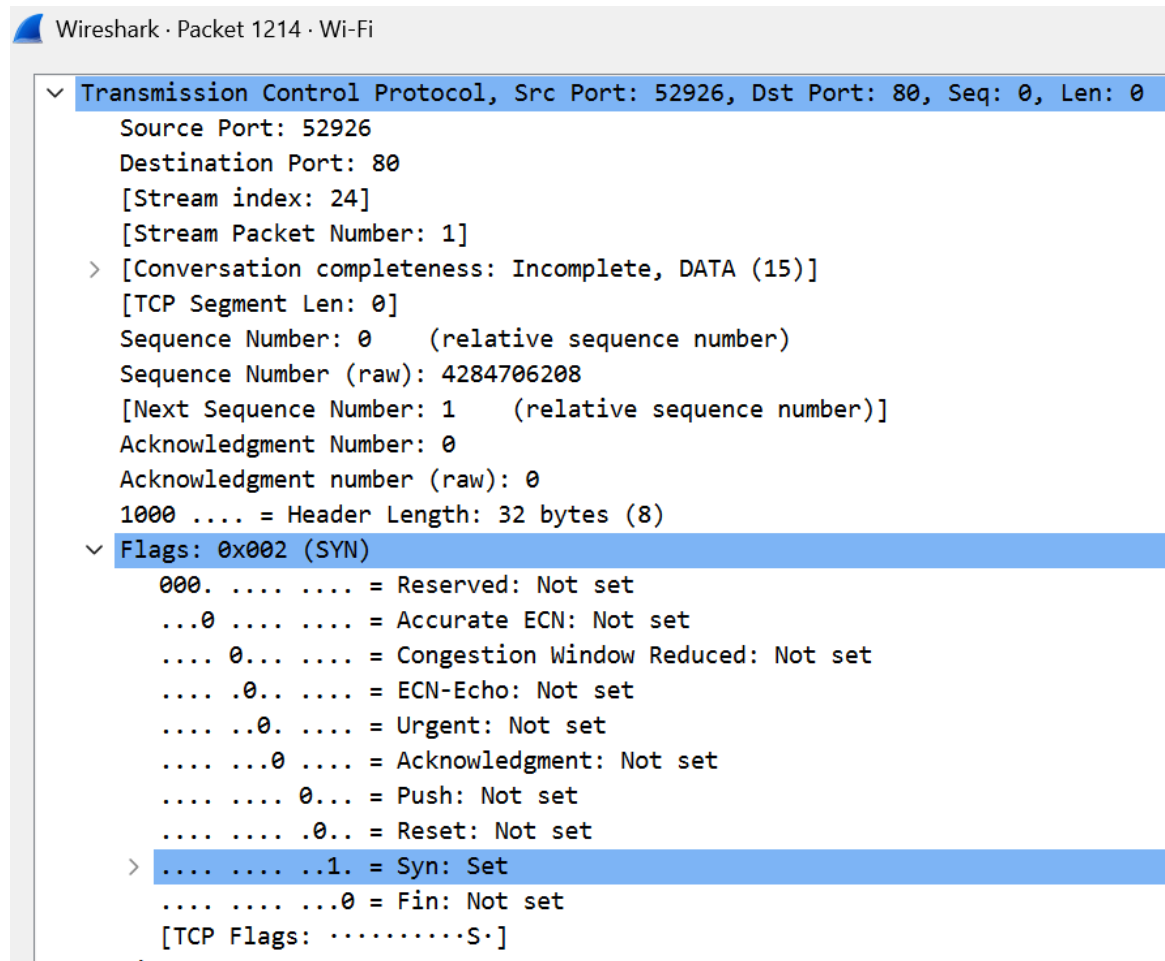
> Frame 368: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Dev
> Ethernet II, Src: WdWinwireless_b2:98:75 (fc:dd:55:b2:98:75), Dst: Intel_29:65:a3 (f8:a
> Internet Protocol Version 6, Src: 2600:1406:3a00:21::173e:2e65, Dst: 2401:4900:2346:4f
> Transmission Control Protocol, Src Port: 80, Dst Port: 52918, Seq: 1, Ack: 1, Len: 0

No.	Time	Source	Destination	Protocol	Length	Info
1214	59.061932	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	86	52926 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1231	59.285030	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	86	80 → 52926 [SYN, ACK] Seq=0 Ack=1 Win=64800 Len=0 MSS=1300 SACK_PERM WS=128
1232	59.285219	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52926 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1234	59.286553	2401:4900:2346:4f31...	2600:1406:3a00:21::...	HTTP	500	GET / HTTP/1.1
1274	59.530624	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52926 [ACK] Seq=1 Ack=427 Win=64384 Len=0
1275	59.530624	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	1052	HTTP/1.1 200 OK (text/html)
1286	59.576401	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52926 → 80 [ACK] Seq=427 Ack=979 Win=64512 Len=0
1307	59.828910	2401:4900:2346:4f31...	2600:1406:3a00:21::...	HTTP	440	GET /favicon.ico HTTP/1.1
1318	60.054603	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	74	80 → 52926 [ACK] Seq=979 Ack=793 Win=64128 Len=0
1320	60.074048	2600:1406:3a00:21::...	2401:4900:2346:4f31...	TCP	1374	80 → 52926 [ACK] Seq=979 Ack=793 Win=64128 Len=1300 [TCP PDU reassembled in 1321]
1321	60.074048	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	422	HTTP/1.1 404 Not Found (text/html)
1322	60.074134	2401:4900:2346:4f31...	2600:1406:3a00:21::...	TCP	74	52926 → 80 [ACK] Seq=793 Ack=2627 Win=65280 Len=0

The first three packets in this newly filtered view is the **TCP 3-way handshake (SYN,SYN-ACK, ACK)** that established the connection for this HTTP request.

Analysis of individual TCP packets

SYN (Initiation)



1. SYN Packet Packet Number : 1214

- **Source Port:** 52926
- **Destination Port:** 80
- **Sequence Number (relative):** 0
- **Sequence Number (raw):** 4284706288
- **Acknowledgement Number (relative):** 0
- **Acknowledgement Number (raw):** 0
- **Flags Set:**
 - **Syn: Set** (0x002)

SYN - ACK (Server's Acknowledgment and Initiation)

```

Transmission Control Protocol, Src Port: 80, Dst Port: 52926, Seq: 0, Ack: 1, Len: 0
  Source Port: 80
  Destination Port: 52926
  [Stream index: 24]
  [Stream Packet Number: 2]
  > [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 0]
  Sequence Number: 0      (relative sequence number)
  Sequence Number (raw): 571335219
  [Next Sequence Number: 1      (relative sequence number)]
  Acknowledgment Number: 1      (relative ack number)
  Acknowledgment number (raw): 4284706209
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x012 (SYN, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    .... 0... = Congestion Window Reduced: Not set
    .... .0.. = ECN-Echo: Not set
    .... ..0. = Urgent: Not set
    .... ...1 = Acknowledgment: Set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
  > .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
  [TCP Flags: .....A..S.]
```

2. SYN-ACK Packet Packet Number : 1231

- **Source Port:** 80
- **Destination Port:** 52926
- **Sequence Number (relative):** 0
- **Sequence Number (raw):** 571335219
- **Acknowledgement Number (relative):** 1
- **Acknowledgement Number (raw):** 4284706289
(This is the client's initial raw sequence number + 1)
- **Flags Set:**
 - **Syn:** Set
 - **Acknowledgment:** Set (0x012)

ACK (Client's Acknowledgment, completing handshake)

```

  Transmission Control Protocol, Src Port: 52926, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
    Source Port: 52926
    Destination Port: 80
    [Stream index: 24]
    [Stream Packet Number: 3]
    > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 1      (relative sequence number)
    Sequence Number (raw): 4284706209
    [Next Sequence Number: 1      (relative sequence number)]
    Acknowledgment Number: 1      (relative ack number)
    Acknowledgment number (raw): 571335220
    0101 ..... = Header Length: 20 bytes (5)
    ✓ Flags: 0x010 (ACK)
      000. .... = Reserved: Not set
      ...0 .... = Accurate ECN: Not set
      .... 0... = Congestion Window Reduced: Not set
      .... .0.. = ECN-Echo: Not set
      .... ..0. = Urgent: Not set
      .... ...1 = Acknowledgment: Set
      .... .... 0... = Push: Not set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: .....A.....]
```

3. ACK Packet Packet Number: 1232

- **Source Port:** 52926
- **Destination Port:** 80
- **Sequence Number (relative):** 1
- **Sequence Number (raw):** 4284706289
(It's now acknowledging the server's SYN-ACK)
- **Acknowledgement Number (relative):** 1
- **Acknowledgement Number (raw):** 571335220
(This is the server's initial raw sequence number + 1, as the client is acknowledging the server's SYN)
- **Flags Set:**
 - **Acknowledgment: Set** (0x010)

Table of TCP 3-Way Handshake Analysis

Packet Type	Source Port	Destination Port	Flags Set	Sequence Number (relative)	Acknowledgment Number (relative)
SYN	52926	80	SYN	0	0
SYN-ACK	80	52926	SYN, ACK	0	1
ACK	52926	80	ACK	1	1

HTTP Analysis

Objective: To identify and extract key information like IP addresses, Host, User-Agent, and Response Code.

http						
No.	Time	Source	Destination	Protocol	Length	Info
370	16.421137	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	584	HTTP/1.0 408 Request Time-out (text/html)
1234	59.286553	2401:4900:2346:4f31...	2600:1406:3a00:21::...	HTTP	500	GET / HTTP/1.1
1275	59.530624	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	1052	HTTP/1.1 200 OK (text/html)
1307	59.828910	2401:4900:2346:4f31...	2600:1406:3a00:21::...	HTTP	440	GET /favicon.ico HTTP/1.1
1321	60.074048	2600:1406:3a00:21::...	2401:4900:2346:4f31...	HTTP	422	HTTP/1.1 404 Not Found (text/html)

1. HTTP GET Request Packet Analysis

```
Wireshark · Packet 1234 · Wi-Fi
> Frame 1234: 500 bytes on wire (4000 bits), 500 bytes captured (4000 bits) on interface \Device\NPF_{45A5D929-A2E7-4B65-9536-D784073213B2}, id 0
> Ethernet II, Src: Intel_29:65:a3 (f8:ac:65:29:65:a3), Dst: WeWinwirele_b2:98:75 (fc:dd:55:b2:98:75)
> Internet Protocol Version 6, Src: 2401:4900:2346:4f31:add2:c616:d126:2c36, Dst: 2600:1406:3a00:21::173e:2e65
> Transmission Control Protocol, Src Port: 52926, Dst Port: 80, Seq: 1, Ack: 1, Len: 426
▼ Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
    Host: example.com\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Response in frame: 1275]
    [Full request URI: http://example.com/]
```

- **Internet Protocol Version 6:**
 - **Source IP Address:** 2401:4900:2346:4f31:add2:c616:d126:2c36
 - **Destination IP Address:** 2600:1406:3a00:21::173e:2e65
- **Transmission Control Protocol:**
 - Src Port: 52926
 - Dst Port: 80
 - Seq: 1, Ack: 1, Len: 426
- **Hypertext Transfer Protocol:**
 - **Request Method:** GET / HTTP/1.1
 - **Host:** example.com
 - **User-Agent:** Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Safari/537.36
 - [Response in frame: 1275] (This helpful Wireshark link points to the corresponding response packet)
 - [Full request URI: http://example.com/]

2. HTTP 200 OK Response Packet Analysis

```
Wireshark · Packet 1275 · Wi-Fi
> Frame 1275: 1052 bytes on wire (8416 bits), 1052 bytes captured (8416 bits) on interface \Device\NPF_{45A5D929-A2E7-4B65-9536-D784073213B2}, id 0
> Ethernet II, Src: WelWinwirele_b2:98:75 (fc:dd:55:b2:98:75), Dst: Intel_29:65:a3 (f8:ac:65:29:65:a3)
> Internet Protocol Version 6, Src: 2600:1406:3a00:21::173e:2e65, Dst: 2401:4900:2346:4f31:add2:c616:d126:2c36
> Transmission Control Protocol, Src Port: 80, Dst Port: 52926, Seq: 1, Ack: 427, Len: 978
▼ Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Accept-Ranges: bytes\r\n
    Content-Type: text/html\r\n
    ETag: "84238dfc8092e5d9c0dac8ef93371a07:1736799080.121134"\r\n
    Last-Modified: Mon, 13 Jan 2025 20:11:20 GMT\r\n
    Vary: Accept-Encoding\r\n
    Content-Encoding: gzip\r\n
    Cache-Control: max-age=1380\r\n
    Date: Wed, 11 Jun 2025 16:58:27 GMT\r\n
  > Content-Length: 648\r\n
  Connection: keep-alive\r\n
  \r\n
  [Request in frame: 1234]
  [Time since request: 0.244071000 seconds]
  [Request URI: /]
  [Full request URI: http://example.com/]
  Content-encoded entity body (gzip): 648 bytes -> 1256 bytes
  File Data: 1256 bytes
```

- **Internet Protocol Version 6:**
 - **Source IP Address:** 2600:1406:3a00:21::173e:2e65
 - **Destination IP Address:** 2401:4900:2346:4f31:add2:c616:d126:2c36
- **Transmission Control Protocol:**
 - Src Port: 80
 - Dst Port: 52926
 - Seq: 1, Ack: 427, Len: 978
- **Hypertext Transfer Protocol:**
 - **Status Line:** HTTP/1.1 200 OK
 - **Response Code:** 200 OK
 - **Content-Type:** text/html
 - **Content-Length:** 648

Detail	GET Request packet (1234)	200 OK Response packet (1275)
Source IP Address	2401:4900:2346:4f31:add2:c616:d126:2c36 (Client)	2600:1406:3a00:21::173e:2e65 (Server)
Destination IP Address	2600:1406:3a00:21::173e:2e65 (Server - IPv6)	2401:4900:2346:4f31:add2:c616:d126:2c36 (Client - IPv6)
Host	example.com	N/A
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Safari/537.36	N/A
Response Code	N/A	200 OK

Conclusion

Demonstrated and analysed how TCP connections are established and how HTTP requests/responses travel over them in a network capture.